Ryley Griffiths

Aspiring graduate software engineer, interested in secure and trustworthy AI.

About me

I am a final-year BSc (Hons) Computer Science student at Teesside University London, graduating in 2027. I am applying for graduate software engineering roles.

My specialist interest is secure and trustworthy AI: how systems that learn or recognise things can be attacked, and how to show they behave as claimed.

Experience

- **Data Intern, London Borough of Hackney** (Future Ready Internship), summer 2026. Built a storytelling application, which is now moving towards deployment.
- **E-commerce and Operations Assistant, Triple Solar**, July 2025 to present.
- **Developer Intern, Paiger**, 2023.

Skills

These are skills used in the projects below and in my roles:

- **Languages:** TypeScript, JavaScript, Python, SQL, HTML and CSS.
- **Web development:** React, Vite, progressive web apps, Flask, and REST APIs with Hono.
- **Cloud and delivery:** Cloudflare Workers and D1, GitHub Actions for CI/CD, GitHub Pages and Render.
- **Security:**
  - authentication design: password stretching and hashed session tokens
  - role-based access control and rate limiting
- **Data and evaluation:** data pipelines, documenting datasets and models with data cards and model cards, and evaluating with held-out test data.
- **Computer vision:** image processing and perceptual hashing for recognising cards on a phone.
- **Ways of working:** automated testing (unit and end-to-end), version control with Git.

Featured projects

- **PokéVault** (TypeScript, React, Cloudflare Workers): a phone web app that recognises Pokémon cards on the device and tracks a collection's value. It has its own API and data pipeline, a data card and a model card, and an evaluation on held-out simulated photos.
- **Enterprise-Challenge** (Python, HTML): a secure web application built for my Enterprise Challenge module and deployed on Render.

- **ORBIT TravelTracker** (Python, JavaScript): a travel tracker built with Flask, SQLite and globe.gl. It shows visited countries on a 3D globe and keeps a wishlist, and friends can overlay their maps using a share code.
- **PokemonShowdownTools** (Python): tools for Pokémon Showdown challenges, such as finding the Pokédex numbers that use the same digits as a given number.
- **Shiny-Tracker-App** (JavaScript): Used in personal fun with Pokemon Shinies

Current investigation

My final-year report investigates timing attacks on neuromorphic vision pipelines, which are built on spiking neural networks. These networks process information as events over time, so the time a pipeline takes to respond could reveal something about what it has seen. I am assessing what this kind of attack would mean for an automotive supplier, as a risk assessment.

Contact

You can reach me through GitHub.